



**ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI**

**Course Structure and Syllabus
(From Academic Session 2018-19 onwards)**

**B.TECH
INDUSTRIAL AND PRODUCTION ENGINEERING
7th Semester**



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Guwahati

Course Structure

(From Academic Session 2018-19 onwards)

B.Tech 7th Semester: Industrial and Production Engineering Semester VII/ B.TECH/IPE

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	IPE181701	Production Planning and Control	3	0	0	3	30	70
2	IPE181702	Tool Design	3	0	0	3	30	70
3	IPE181703	Industrial Engineering and Operations Research	3	0	0	3	30	70
4	IPE1817PE1*	Program Elective -1	3	0	0	3	30	70
5	IPE1817OE1*	Open Elective -1	3	0	0	3	30	70
6	HS181704	Principles of Management	3	0	0	3	30	70
Practical								
1	IPE181722	Project-1	0	0	8	4	50	50
2	IPE181723	Grand Viva Voce-I	0	0	0	1	-	50
3	SI181721	Internship-III (SAI-Industry)	0	0	0	2	-	200
TOTAL			18	0	8	25	230	720
Total Contact Hours per week: 26								
Total Credit: 25								

Program Elective-1		
Sl. No.	Subject Code	Subject
1	IPE1817PE11	Environment Engineering and Renewable Energy Sources
2	IPE1817PE12	Supply Chain Management
3	IPE1817PE13	Project Planning and Appraisal
4	IPE1817PE14	Micro Manufacturing Processes and Systems
5	IPE1817PE15	Plant and Maintenance Engineering
6	IPE1817PE16	Manufacturing System Simulation
7	IPE1817PE1*	Any other subject offered from time to time with the approval of the University

Open Elective-1		
Sl. No.	Subject Code	Subject
1	IPE1817OE11	Total Quality Management
2	IPE1817OE12	Production Economics and Financial Management
3	IPE1817OE13	CNC Machines and Robotics
4	IPE1817OE1*	Any other subject offered from time to time with the approval of the University

Detailed Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE181701	Production Planning and Control	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of production, planning and control in real life situations and analyze economic analysis

CO2: To design the better process design in practice.

CO3: To identify and apply the scheduling, dispatching and network logics.

CO4: To apply concept of sale forecasting in real life situations.

MODULE 1: The Need for Production Planning and Control (PPC)

Functions – Production planning versus Production planning versus Production Control. Factors influencing PPC: Project and Jobbing production, Batch, Mass and Flow production, Continuous production – Planning for meeting seasonal or occasional increase in demand. Centralized and decentralized PPC.

MODULE 2: Pre-requisites of PPC

Product design and analysis – marketing, functional and operational analysis; Data for raw materials, Equipment and Tooling. Economic Analysis: Profit and competitiveness – standardization and Simplification – Economic lot size.

MODULE 3: Process Design

Product planning or buy decision, value analysis. Process planning steps and process sheets – Routing – Tool control. Cost benefit and Break-even analysis. Process planning in different situations.

MODULE 4: Scheduling

Concepts – Factors influencing scheduling: Master Schedule. Job Schedules – Schedule techniques – Gantt Chart. Mathematical loading and scheduling – Index method – Sequencing. Project Scheduling – Network logic and Steps in the use of critical Path Analysis.

MODULE 5: Dispatching

Introduction and Functions – How functions are performed – Dispatching under different situations – Documents in Dispatching.

MODULE 6: Sales Forecasting

Need and meaning – Forecasting techniques – market survey; Experimental smoothing; Regression analysis, sessional variations. Extrapolating Future Demand.

Textbooks/Reference Books:

1. Production Planning and Control (Everest Publisher) ----- L. C. Jhamb.
2. Elements of PPC (Universal Book) ----- S. Elton.
3. Production Systems, Planning analysis and Control ----- J. L. Riggs.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE181702	Tool Design	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of selection of tool materials in tool designing.

CO2: To design various tools for different purposes.

CO3: To identify and design the various jigs and fixtures for specific machining requirements.

CO4: To identify and design the various press tools and dies for specific machining requirements.

MODULE 1: Tool Design

Tool design consideration, Selection of tool materials, Tooling economics and safety as related to tool design. Design of Single point cutting tools, Design of Carbide and Ceramic tipped tools, Design of Chip breakers. Design of Multi point cutting tools, Design of Broaches, Twist drill, Reamers and Milling cutters.

MODULE 2: Introduction: Classes of Engineering Production

Definition and functions of Jigs and fixtures. Principles of location – Locator Types, clamping principles – Types of clamp: Drill bushes and plate – bush types and materials. Design considerations.

MODULE 3: Design of Drill Jigs

Types, Jigs and Machine relationship. Jig body and feet. Examples of typical drill Jigs. Design for specific product.

MODULE 4: Design of Fixtures

Types, Fixture and machine relationship – Examples of typical milling fixtures – Design. Turning, Grinding and broaching fixtures.

MODULE 5: Press Tools

Types of Presses; Press Tool operations – Computation of capacities and tonnage for blanking, piercing, bending, forming and drawing operations. Principle and design procedure for press tools. Detailed designing procedure for benching and planing operations examples.

MODULE 6: Dies

Types – Combination and progressive die; Single, double and triple action dies. Detailed design procedure for blanking and piercing dies with examples.

Textbooks/Reference Books:

1. Production Tooling Equipment (B.I. Publication) ----- S.A.J. Parsons.
2. Fundamentals of Tool design (PHI) ----- A S T M E.
3. Jigs and Fixture (Jain Brothers) ----- Griant.
4. Jigs and Fixture (TATA McGraw Hill) ----- Joshi
5. Press Tool : Design and Calculation (A.H. Wheeler % Co.) ----- P. H. Joshi
6. Geofferey B. and Winston K.A., ' Fundamentals of Machining and Machine tools', Published by Marcel Dekker, Inc [ISBN-08247-7852-9] 1989
7. Kumar S. & Chandra U, 'Production engineering Design (tool Design) Published by satya Prakashan [ISBN-81-7684-495-0-] 2008

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE181703	Industrial Engineering and Operations Research	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the industrial engineering concept in real life situations.

CO2: To develop the knowledge on job analysis, method development and work measurement for reduction in time and effort in industrial situation.

CO3: To apply concept of product/process design and reliability in new product development.

CO4: To formulate mathematical models for optimization of engineering & business problems for quantitative analysis, solution and interpretation of results.

MODULE 1: Introduction

Definition and objectives of Industrial Engineering, Scope of Industrial Engineering, Production systems and their classifications; Productivity-Total and partial productivity, Reasons and remedy for poor productivity.

MODULE 2: Job Analysis and Work Measurement Systems

Work System Design: Taylor's scientific management, Gilbreth's contributions; method study, micro-motion study, principles of motion economy; work measurement - stop watch time study, micro motion and memo motion, work sampling, standard data, PMTS; ergonomics; job evaluation, merit rating, incentive schemes, and wage administration; business process reengineering

MODULE 3: Product and Process Design

Product design and development processes, product life cycle, Process flow chart, Types of processes, Process performance, Learning curve.

MODULE 4: Reliability and Maintenance

Reliability, availability and maintainability; distribution of failure and repair times; determination of MTBF and MTTR, reliability models; system reliability determination; Maintenance management and its objectives, Various types of Maintenance Planning, House Keeping, 5S concepts.

MODULE 5: Introduction

(a) History of Operations Research

(b) Engineering applications

General Statement of an OR problem

MODULE 6: Linear Programming

(a) Deterministic Linear Programming Problems (LPP): Problem formulation, Feasibility and Optimality, Basic and Non-Basic solutions

(b) Solution of LPP by

(i) Graphical method: (1) Corner point method (2) Iso-profit (cost) method

(ii) Simplex Algorithm: Use of Slack, Surplus and Artificial variables and their meanings, Condition for starting solution and limitation of simplex method, Meaning and significance of Big-M (Charnes method of penalty) and 2-phase method

(iii) Meaning and examples of Unique, Alternate/Multiple, Unbounded and Infeasible solutions, Degeneracy and Cycling

MODULE 7: Special LPPs

(a) Integer Programming Problem (IPP) and solution by Simplex and Dual Simplex algorithms

(b) Transportation Problem (TP): Solution steps as (i) Initial Basic Feasible Solution (IBFS), (ii) Check for degeneracy and (iii) Moving towards optimality

Assignment Problem (AP) and Degeneracy. Discussion on method extended to Travelling Salesman Problem (TSP).

Textbooks/Reference Books:

1. Heizer J. and Render B., "Operations Management" published by Pearson Education Ltd, 11th ed., ISBN : 978-0273787071, 2013,
2. Chase R.B., Aquilano J., Jacobs F., "Modern Production/Operations Management" published by Richard D Irwin ISBN-139788126513727 , 2007
3. Sharma J.K., "Operation's Research" published by MacMillan Publishers India Ltd., ISBN - 978-9350-59336-3, 2013
4. Barnes Ralph M., "Motion & Time study: Design and Measurement of Work", Wiley Text Books, **ISBN-10:** 8126522178, 2009.
5. Operations Research – H A Taha
6. Operations Research – Gupta and Hira
7. Operations Research – Billy E Gillet

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE11	Environment Engineering and Renewable Energy Sources	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of ecological engineering in real life environmental engineering problems.

CO2: To implement the various pollution control guidelines on different kinds of pollutions and design control devices.

CO3: To identify various types of energy systems and design devices to store it.

CO4: To develop concept on devices for alternate energy sources.

MODULE 1: Earth's Natural Resources

Soil, Water, Forest, Wildlife etc., Renewable and non-renewable resources – Finite nature of the natural resources – basic human requirements – overexploitation of the resources; towards sustainable developments.

MODULE 2: Environmental Pollution

Effects of Pollutants on living systems – Solid waste management. Global Environment concerns, Global warming- causes and effect, Climate change and response & Clean Development Mechanism, Kyoto Protocol, Carbon trading.

MODULE 3: Environment and Future

Pollution growth – regulation; Modern Technology and Population explosion; World Food Production – population growth in India – Green revolution. Growth and Urbanization: factors contributing to urbanization – urban environment – health housing, Education etc. – management of urbanization.

Environmental Impact Assessment process – Goals, Objectives, Survey, Prediction and analysis; alternative plans – implementation and monitoring. Cost benefit analysis.

Environment and sustainable developments with regard to population growth, Urbanization and Industrial growth –Threat to sustainable development – development project and nature management.

MODULE 4: Solar and Wind Energy

Solar radiation- beam and diffuse radiation, flat plate collectors, concentrating collectors, solar water heaters, solar cooker, storage of solar energy-thermal storage, photovoltaics'- solar cells & its application.

Principle of wind energy conversion – basic components, vertical and horizontal axis wind mills, wind data and site selection considerations.

MODULE 5: Energy from Biomass

Biomass conversion technologies - Biogas generation plants, classification, site selection – Biomass Gasification process and gasification plants, ethanol production, bio-diesel production. Biomass densification and briquette.

MODULE 6: Geothermal and Ocean Energy

Geothermal sources and resources like hydrothermal, geo-pressurized hot dry rock, magma, advantages and disadvantages and application of geothermal sources.

Ocean thermal electric conversion (OTEC) systems like open cycle, closed cycle, hybrid cycle.

MODULE 7: Fuel Cell

Introduction, Design principle and operation of fuel cell, Types of fuel cell and application.

Hydrogen energy – Production methods, Hydrogen storage, hydrogen transportation, hydrogen as alternative fuel for vehicles

Textbooks/Reference Books:

1. Environmental Engineering by Gerard Kiely, TMH Education Pvt. Ltd.
2. Non-conventional energy sources by G.D. Rai, Khanna Publishers.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE12	Supply Chain Management	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of Supply Chain Management to solve problem based business issues.

CO2: To implement the various techniques of logistics, purchasing & vendor management to make correct decisions.

CO3: To identify and apply the systems of inventory management in practice.

CO4: To apply concept of computer / IT in Supply Chain Management

MODULE 1: Introduction

Basic Concept & Philosophy of Supply Chain Management; Essential features, various flows (cash, value and information), Key Issues in SCM, benefits and case examples.

MODULE 2: Logistics Management

Logistics as part of SCM, Logistics costs, different models, logistics sub-system, inbound and outbound logistics, bullwhip effect in logistics, Distribution and warehousing management.

MODULE 3: Purchasing & Vendor Management

Centralized and Decentralized purchasing, functions of purchase department and purchase policies. Use of mathematical model for vendor rating / evaluation, single vendor concept, management of stores, accounting for materials.

MODULE 4: Inventory Management

Concept, various costs associated with inventory, various EOQ models, buffer stock (trade off between stock out / working capital cost), lead time reduction, re-order point / re-order level fixation, exercises –numerical problem solving, ABC, SDE / VED Analysis, Just-In-Time & Kanban System of Inventory management.

MODULE 5: Recent Issues in SCM

Role of Computer / IT in Supply Chain Management, CRM Vs SCM, Benchmarking-concept, Features and Implementation, Outsourcing-basic concept, Value Addition in SCM-concept of demand chain management.

Textbooks/Reference Books:

1. G. Raghuram (I.I.M.A.) - Logistics and Supply Chain Management (Macmillan, 1st Ed.)
2. Dr. Gopal Krishnan - Material Management, (Pearson, New Delhi, 5th Ed.)
3. D.K. Agarwal. - A Text Book of Logistics and Supply chain management (Macmillan, 1st Ed.).
4. B.S. Sahay - Supply Chain Management (Macmillan, 1st Ed.)
5. Sunil Chopra and Peter Meindl - Supply chain management (Pearson, 3rd Ed.)

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE13	Project Planning and Appraisal	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the project management philosophy in real projects.

CO2: To implement the various Project appraisal methods in practice.

CO3: To identify and analyze the financial analysis and project finance along with analysis of Project Networks in industry applications.

CO4: To apply concept of project monitoring and control in real life projects.

MODULE 1: Introduction

Concept of Project – Characteristics and classification – aspect of project: Project identification – selection criteria and feasibility analysis. Project Formulation – Steps – Planning and Evaluation.

MODULE 2: Project Appraisal

Concept and scope – stages in project appraisal – appraisal criteria and Methodology.

MODULE 3: Financial Analysis and Project Finance

significance of Financial analysis; Financial Tools – preparation of fund flow statement, Cash flow measurement; ratio analysis – Advantages and limitations – Break-Even analysis. Project Financing – Sources.

MODULE 4: Analysis of Project Networks

PERT and CPM -- Network representation – Rules to setup Networks – Analysis. Structuring of PERT Data – Arrow diagram, Work break down structure.

MODULE 5: Project Monitoring and Control Aspects

Project Management under risk and uncertainty – using computer for project management. Towards better project management – Bottlenecks and remedies.

Textbooks/ Reference Books:

1. Project Management (Himalaya Publishing house) ----- V. Desai.
2. Project Management (Code No. 129, AICTE) ----- K. C. Sahu.
3. Project Management with PERT, CPM etc. (Jain Brothers) ---- J.C. Pant.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE14	Micro Manufacturing Processes and Systems	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of Micro-Manufacturing processes to design Precision Engineering products.

CO2: To identify and implement the error control processes to control error sources.

CO3: To implement the various Micro machining processes in practical applications to solve engineering problems.

CO4: To apply concept of MEMS in automobile, defense, health care, Industrial, aerospace etc.

MODULE 1: Micro Manufacturing

Introduction to micro-manufacturing: definition, need/importance, applications. Size effect. Classification of micro-manufacturing processes. Introduction – Precision, Accuracy & Smoothness – Need – Development of overall machining precision- Classes of achievable machining Accuracy- Precision Machining-High Precision Machining-Ultra Precision Machining-application of precision machining- Materials for tools and machine elements – carbides – ceramic, CBN & Diamond-Tool and work material compatibility.

MODULE 2: Precision Machine Element

Introduction – Guide ways – Drive systems – Spindle drive – preferred numbers - Rolling 83 elements – hydrodynamic & hydrostatic bearings –Hybrid fluid bearings- Aero static and aero dynamic Bearings-Hybrid gas bearings-materials for bearings.

MODULE 3: Error Control

Error – Sources – Static stiffness – Variation of the cutting force – total compliance – Different machining methods – Thermal effects – heat source – heat dissipation – Stabilization – decreasing thermal effects – forced vibration on accuracy – clamping & setting errors – Control – errors due to locations – principle of constant location surfaces.

MODULE 4: Micro Manufacturing Processes

Micro machining processes-diamond machining - micro engraving - Micro replication techniques-formingcasting- injection moulding - micro embossing - Energy assisted processes - LBM, EBM, FIB, Micro electro discharge machining photolithography- LIGA process- Silicon micro machining- Wet and dry etching-thin film deposition. Molecular dynamics at atomistic scale, focused ion-beam techniques, abrasive micro-finishing techniques. Microforming techniques: laser micro-bending, micro-deep drawing and micro-extrusion. Microwelding and joining techniques. Micro-fabrication using deposition techniques such as epitaxial, sputtering, chemical vapor deposition (CVD) techniques Sensors and actuators for micro-manufacturing. Metrology for micromanufacturing. Introduction to nano-scale manufacturing.

MODULE 5: MEMS Introduction

MEMS –characteristics- principle – Design – Application: automobile, defense, health care, Industrial, aerospace etc.

Textbooks/Reference Books:

1. Venkatesh V.C. and Izman S., “Precision Engineering”, Tata McGraw Hill, ISBN:9780070620902, 2007.
2. Murthy R.L., “Precision Engineering”, New Age International, ISBN: 9788122407501, 2009.
3. Nakazawa H., “Principles of Precision Engineering”, Oxford University Press, ISBN 10: 0198562667 1994.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE15	Plant and Maintenance Engineering	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the knowledge of maintenance planning, economics, scheduling and Computer-aided maintenance management system in practical life.

CO2: To implement the right maintenance system out of various Maintenance categories by knowing its merits and demerits to maintain quality of maintenance.

CO3: To identify the right process by Condition Monitoring and analyze Cost comparison with and without condition monitoring.

CO4: To apply concept of repair methods for machine elements and material handling equipment in practice.

MODULE 1: Introduction

Emphasizing the functional similarity as a conversion medium of different types of plants; Common features in terms of infrastructural input requirements; some typical plant, and their system configurations. Climatological, economic, social legal, political, industrial determinants of site selection; Establishing various kinds of weighing coefficients of these factors in relation to the nature of the plant. Techno-economic viability study, Depreciation, LCC, Replacement analysis.

MODULE 2: Principles and Practices of Maintenance Planning

Basic Principles of maintenance planning – Objectives and principles of planned maintenance activity – Importance and benefits of sound Maintenance systems – Reliability and machine availability – MTBF, MTTR and MWT – Factors of availability – Maintenance organization – Maintenance economics. Maintenance planning and scheduling. Maintenance organization – objectives and characteristics – centralized and decentralized maintenance. Maintenance costs – classification of maintenance costs – maintenance cost analysis – cost effectiveness analysis. Maintenance budgeting – types of maintenance budget – preparation of maintenance budget. Human factor in maintenance – manpower planning for maintenance – objectives and stages of manpower planning – training for maintenance personnel. Computer-aided maintenance management system (CMMS) – functions, applications and advantages of CMMS. Maintenance integration – various steps in integration – scheme of integration of maintenance function with other functions.

MODULE 3: Maintenance Policies – Preventive Maintenance

Maintenance categories – Comparative merits of each category – Preventive maintenance, maintenance schedules, repair cycle - Principles and methods of lubrication – TPM. Total productive maintenance (TPM) – features – methodology – basic systems of TPM – TPM and tero-technology. Six sigma maintenance. Lean maintenance – 5-zero maintenance concept – 5-S maintenance concept. Business centered maintenance (BCM) – six pillars – success factors. Maintenance effectiveness – overall equipment effectiveness – key performance indicators – maintenance performance measuring indices. Quality assured maintenance – need – maintenance work quality – use of c-chart for quality control in maintenance.

MODULE 4: Condition Monitoring

Condition Monitoring – Cost comparison with and without CM – On-load testing and offload testing – Methods and instruments for CM – Temperature sensitive tapes – Pistol thermometers – wear-debris analysis. Vibration monitoring – vibration fundamentals – vibration analysis. Vibration transducers – types. Machinery vibration trouble shooting – machinery vibration standard, severity chart and acceptable limits. Lubricant monitoring – components and techniques – filter debris analysis & filter grams. Crack monitoring – techniques. Corrosion monitoring – techniques.

MODULE 5: Repair Methods for Basic Machine Elements

Repair methods for beds, slide ways, spindles, gears, lead screws and bearings – Failure analysis – Failures and their development – Logical fault location methods – Sequential fault location. Reliability centered maintenance (RCM) – steps – flow diagram – basic guidelines. Fault tree analysis (FTA), Event tree analysis (ETA), Root cause analysis (RCA), Failure modes and effects analysis (FMEA), Failure mode effect criticality analysis (FMECA)

MODULE 6: Repair Methods for Material Handling Equipment

Repair methods for Material handling equipment - Equipment records –Job order systems -Use of computers in maintenance.

Textbooks/Reference Books:

1. Ebeling Charles E., "An introduction to Reliability and Maintainability Engineering", Tata McGraw-Hill Publishing Co. Ltd., New Delhi, [ISBN 10 0070421382], 2000.
2. Srinath, L.S., "Reliability Engineering", Affiliated East –West Press Ltd., New Delhi, [ISBN 10 8176710482], 2011
3. Dhillon, B.S., "Engineering Maintainability", Prentice Hall of India, New Delhi, 2000.
4. Blanchard, Benjamin, S., "Logistics Engineering and Management", Pearson, [ISBN 10 1292027134], 2013.
5. Gupta A. K., Reliability, Maintenance and Safety Engineering, University Science Press, New Delhi, 2009.
6. Rao S. S., Reliability-Based Design, McGraw-Hill, Inc, New York, 1992.
7. Srivastava S. K., Maintenance Engineering and Management, S. Chand & Company Ltd., New Delhi, 1998.
8. Venkataraman, Maintenance Engineering and Management, Prentice-Hall of India Pvt. Ltd., New Delhi, 2007.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817PE16	Manufacturing System Simulation	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the concept of Probability and Statistics to design Simulation experiments.

CO2: To design the various simulation experiments and analyze their outputs, Interpretation of data and Validation of data in industrial applications.

CO3: To identify design features and requirements with regard to assembly, product design for manufacturing.

CO4: To apply concept of simulation techniques to design simulation models, simulation of translational and rotational mechanical systems, simulation of hydraulic systems.

MODULE 1: Introduction

Basic Concept of System – Elements of Manufacturing System – Concept of Simulation – Simulation as A Decision Making Tool – Types of Simulation – System Modeling – Types of Modeling.

MODULE 2: Random Numbers

Probability and Statistical Concepts of Simulation – Pseudo Random Numbers – Methods of Generating Random Numbers – Discrete and Continuous Distribution – Testing of Random Numbers – Sampling – Simple, Random and Simulated.

MODULE 3: Design of Simulation Experiments

Problem Formulation – Data Collection and Reduction – Time Flow Mechanical – Key Variables – Logic Flow Chart Starting Condition – Run Size – Experimental Design Consideration – Output Analysis, Interpretation and Validation – Application of Simulation in Engineering Industry.

MODULE 4: Simulation Language

Study of GPSS (Basic Blocks Only) Generate, Queue, Depart, Size, Release, Advance, Terminate, Transfer, Enter and Leave.

MODULE 5: Case Studies

Development of Simulation Models Using GPSS for Queuing, Production, Inventory, Maintenance and Replacement Systems, (Students May Be Asked to Prepare and Present the Case Studies)

MODULE 6: Design for Manufacture and Assembly

Design for Manufacture and Assembly (DFMA). Reasons for not implementing DFMA. Advantages of DFMA with case studies. Design features and requirements with regard to assembly, product Design for Manufacture in relation to any two manufacturing processes: machining and injection molding. Need, objectives.

MODULE 7: System Simulation

Techniques of simulation, Monte Carlo method, Experimental nature of simulation, Numerical computation techniques, Continuous system models, Analog and Hybrid simulation, Feedback systems, Computers in simulation studies, Simulation software packages. Simulation of Mechanical Systems: Building of Simulation models, Simulation of translational and rotational mechanical systems, Simulation of hydraulic systems.

Textbooks/Reference Books:

1. Product Design and Development, “Karl T. Ulrich, Steven D. Eppinger”Mc GrawHill.ISBN:9780072296471
2. Integrated Product and Process Development, “John M. Usher, Utpal Roy and H. R. Parasaei.ISBN: 978-0-471-15597-3
3. Product Design for Manufacture and Assembly, “G. Boothroyd, P. Dewhurst and W. Knight” MarceDaker.ISBN:978-1420089271
4. Engineering Design and Design for Manufacturing: A structured approach, “John R. Dixon and CPoli” Field Stone Publishers, USA. ISBN: 9780964527201
5. Material Selection in Mechanical Design, “M. F. Ashby”Elsevier. ISBN: 9780080419077
6. Jerry Banks And John S.Carson, “Discrete Event System Simulation”, Prentice Hall 1991.
7. John H.Mize And J.Grady Cox, “Essential Of Simulation” – Prentice Hall 1989.
8. Geoffrey Gordon “System Simulation” – Prentice Hall Of India, 1992
9. Jeffrey L.Written, Lonnie D, Bentley And V.M. Barice, “System Analysis And Design Methods”, Galgotia Publication, 1995
10. Averill M.Law And W.David Kelton, “Simulation Modelling And Analysis”, McGraw Hill International Editions, 1991
11. Shannon R.E., “System Simulation”, Prentice Hall 1993.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817OE11	Total Quality Management	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the philosophy of TQM in quality management.

CO2: To justify the various dimensions of product quality and service quality for different product or service types.

CO3: To identify the causes and sub causes of processes by applying the various quality tools and techniques.

CO4: To analyze the customer requirements and technical requirements to design better quality system.

MODULE 1: Introduction

Introduction to concept of quality- Need for quality – Evolution of quality – Definition of quality – Dimensions of manufacturing and service quality – Basic concepts of TQM – Definition of TQM – TQM Framework – Contributions of Deming, Juran and Crosby – Barriers to TQM.

MODULE 2: TQM Principles

Leadership – Strategic quality planning and Quality statements

Customer focus –Customer orientation, Customer satisfaction, Customer complaints, Customer retention

Employee involvement– Motivation, Empowerment, Team and Teamwork, Recognition and Reward, Performance appraisal

Continuous process improvement – PDSA cycle, 5s, Kaizen

Supplier partnership – Partnering, Supplier selection, Supplier Rating.

MODULE 3: TQM Tools and Techniques

The seven traditional tools of quality.

Six-sigma: Concepts, methodology, applications to manufacturing and service sector.

Bench marking– Reason to bench mark, Bench marking process

FMEA – Stages, Types.

Quality circles.

Quality Function Deployment (QFD).

Taguchi's quality engineering: Quality defined by Taguchi and loss function, Taguchi's design of experiments.

Cost of Quality and Performance measures.

TPM Concepts and improvement needs in organization.

MODULE 4: Quality Systems

Meaning of quality system and its present need.

ISO 9000, QS 9000 and ISO 14000: Basic concepts, requirements and benefits.

Textbooks/Reference Books:

1. Dale H. Besterfield, at., "Total Quality Management", Pearson Education India
2. Poornima M Chandrimath, "Total Quality Management", Pearson Education India
3. R K Roy, "Design of Experiments Using the Taguchi Approach", John Wiley & sons
4. M Gopalakrishnan, "Six sigma", PHI
5. Surash Dalela, "ISO 9000: a manual for TQM", S Chand & Co.
6. Girish Pathak, "Total Quality Management- Macmillan publishers India Ltd.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817OE12	Production Economics and Financial Management	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the knowledge of economic theory and production management decision making in practical field.

CO2: To implement the knowledge of construction analysis, interpretation of standard financial statements, dividend policy and share valuation to take better decisions.

CO3: To identify and apply the right financial planning policy to maintain minimum risk for the firm.

CO4: To apply concept of financial analysis and capital budgeting in practice.

MODULE 1: Production Economics

Interaction between economics theory and production and management decision making, concept of firm, industry and economy. Consumer behavior utility, indifference curves and maps, consumers' surplus, demand determinations, demand function. Production function, economy of scale, effect of technology, equilibrium condition, optimal expansion path, short and long run cost functions. Perfect competition and monopoly, equilibrium of firm and industries, different models of firm. Determination of price, price discrimination, pricing of joint products.

MODULE 2: Duality Theory

Definitions and assumptions, Cost functions, Profit functions, Empirical considerations, Profit maximization, Cost minimization.

MODULE 3: Fundamentals of Financial Management

Introduction, Objectives, goals and functions of financial Management. Time value of Money, Risks and Returns, Valuation of Securities.

MODULE 4: Financial Instruments and Markets

Forms of Business Organizations, capital structure, Construction analysis and interpretation of standard financial statements, Dividend policy and share valuation.

MODULE 5: Capital Budgeting

Concept of Operating cycles and working of Capital Management; Capital Budgeting: Cost of capital investment decisions and Risk analysis in Capital Budgeting.

MODULE 6: Financial Planning

Planning of profit and leverages (operating and financial) Risk and Return: Nature of risk, Inter-relationship in risk and return, effect of Tax on return. Cost Accounting-Definition and scope of cost accountancy and costing methods, elements of cost overhead classification distribution and absorption, process costing- Uniform, Marginal and standard costing methods, Case studies

MODULE 7: Inflation and Financial Management

Inflation and Asset revaluation, firm value, financial market returns, financial analysis, capital budgeting.

Textbooks/ Reference Books:

1. Khan and Jain, Financial Management, Tata McGraw Hill, [ISBN-10 007106785X, 13: 9780070656147], 2012
2. Pandey M.I, Financial management, Vikas Publishing House, Tenth edition,[ISBN-10 : 8125937145],2010.
3. Beattie, B.R., and C.R. Taylor. The Economics of Production. John Wiley & Sons, Inc. New York, NY. 1985. (Second edition, 2009, is also okay)
4. Chambers, R.G. Applied Production Analysis: A Dual Approach. Cambridge University Press. Cambridge. 1988.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1817OE13	CNC Machines and Robotics	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

CO1: To apply the numerical control (NC) systems in real life CAM.

CO2: To develop the various NC programming and identify the adaptive control processes.

CO3: To identify and apply the fundamentals of robot, its control system and sensing system.

CO4: To construct mechanical design of robots.

MODULE 1: Concepts and Features of NC Systems

Classification of NC systems - Design considerations of NC machine tools - Constructional features of CNC machine tools. Basic principles of numerical control; Methods of coding Computer Numerical Control (CNC) System, Machine Structure, drive system, motion elements. DNC systems.

MODULE 2: Functions of MCU

Machining center - Turning center – CNC EDM, Ball screws, Bearings, Centralized lubrication systems. Manual part programming – Preparatory, Miscellaneous functions

MODULE 3: Sinumeric, Fanuc Controls

Computer aided part programming – Post processors - APT programming-CNC programming based on CAD Feedback devices.

MODULE 4: Tooling for CNC Machine

Interpolators. Point-to-point and contouring systems

MODULE 5: Adaptive Control

ACO and ACC systems. Maintenance of CNC Machines- Economics of manufacturing using CNC machines

MODULE 6: Introduction to Robotics

Classification of Robots, Characteristics of Robots, performance, advantages and disadvantages of a Robot, industrial applications of a Robot.

MODULE 7: Fundamentals of a Robot

Various system, structure and definition, terms relating to industrial Robots, basic terms related to Robot performance and Characteristics, Control volume of a Robot. Overview of mobile robotics, applications. Sensors and estimation. Distributed robotics. Overview of MEMS, scaling effects, micromanipulation. Microscope optics, depth from defocus, focus measures. Examples from current research.

MODULE 8: Robot languages and programming.

MODULE 9: Controlling the Robot Systems

Introduction to drives, Mechanical, Hydraulic, Pneumatic, electric drives, feedback control.

MODULE 10: Sensing System for a Robot

Introduction, types of sensors, machine vision, Artificial intelligence, Control techniques. Elements of a vision system, lighting, sensors, optics. Geometry of imaging, projections, distortions, depth of field. Digitization, brightness, color space, color depth, image formats. Camera calibration.

MODULE 11: Robot Safety

Introduction, potential safety hazards, safety planning check lists, safety guidelines, latest development in safety measurement.

MODULE 12: Mechanical Design of Robots

Sensors, actuators, gearboxes, robot end-effectors, resolution, accuracy, precision. Describing the position and orientation of objects in 3D space. Coordinate frames, position, orientation and velocity vectors in 3D, coordinate transformations. Applies directly to Computer Graphics. How to mathematically define a path in space. How to control the robot to follow that path Contact tasks, force sensing and control? Also: Haptic interfaces parallel kinematics. Binary images, thresholding, histograms. Area/moment statistics, morphological operations. Segmentation, blob analysis, labeling. Spatial operations and transformations: Pixel neighborhoods, convolution. Mean, Gaussian, Laplacian, gradient filters. Edge detection, Canny, Hough transform.

Textbooks/Reference Books:

1. Craig J John, Introduction to Robotics: Mechanics and Control, , Pearson education,[ISBN-10 0201543613], 2003.
2. Y.Koren , Robotics for Engineers, , McGraw Hill Publications, [ISBN-10 0070353999],1985.
3. Koren, Y. “Computer Control of Manufacturing Systems”, McGraw Hill Book co. New Delhi, 1986. ISBN-10 0070607435
4. Kundra T. K., Rao P. N., and Tiwari N. K., “CNC and Computer Aided Manufacturing”, Tata McGraw Hill, New Delhi, 1991. 3.Fitzpatric,M. “ Machining And CNC Technology” , McGraw-Hill College, 2004, ISBN 10: 0074631039
5. Radhakrishnan P., “Computer Numerical Control Machines”, New Central Book Agency, Calcutta, 1992 ISBN-10 8122433979

Course Code	Course Title	Hours per week L-T-P	Credit C
HS181704	Principles of Management	3-0-0	3

MODULE 1: Introduction

Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc., principles of management, Difference between administration and management

MODULE 2: Financial Management

Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

MODULE 3: Marginal costing

Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

MODULE 4: Cost Accounting

Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

MODULE 5: Capitalisation

Definition and meaning of capitalisation, over and under capitalisation.

MODULE 6: Motivation

Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

MODULE 7: Leadership

Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership.

Textbooks/Reference Books:

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajvive
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharj

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE181722	Project-1	0-0-8	4
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			

Course Code	Course Title	Hours per week L-T-P	Credit C
SI181721	Internship-III (SAI - Industry)	0-0-0	2
GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME			
